Restricted - مقيد

الجامعة السعودية الإلكترونية
SAUDI ELECTRONIC UNIVERSITY
2011-1432

College of Computing and Informatics

Fundamentals of Database

IT403

Project

Deadline: Monday 09/08/2026 @ 23:59

[Total Mark is 20]

Name: Mohammed Alsaleem **ID: S220012336**
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Instructions:

- You must submit two separate copies **(one Word file and one PDF file)** using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets, convert text into image or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You MUST show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

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Task 4 – Reflection (2 Marks)

Project Instructions

- You can work on this project as a group (minimum 3 and maximum 4 students). Each group member must submit the project individually with all group member names mentioned on the cover page.
- This project **worth 20 marks** and will be distributed as in the following:
  - Database Design. **(6 marks)**
  - Database Implementation. **(6 marks)**
  - SQL Operations. **(6 marks)**
  - Project Reflection. **(2 marks)**
- Each leader in the group must submit one report about their chosen Project via the Blackboard **(Email submission will not be accepted and will be awarded ZERO marks)** containing the following:
  A. Task 1 : database design
  B. Task 2: database implementation
  C. All requested queries/results
  D. Screenshots from MySQL (or any other software you use) of **all the tables** after population and **query results**.
  E. Reflection (300–500 words) describing the team's experience, challenges, design decisions, and AI usage (if applicable).
- Submit one **SQL file** containing the complete database implementation, including database creation, tables, data insertion, required queries, VIEW, and TRIGGER.
- Submit a **GitHub** (or GitLab) repository containing the complete project with regular commits and meaningful commit messages.
- Submit a **progress report** describing the completed work, key decisions, challenges, future plan, and contribution of each group member.
- Include **links** to the live report document, GitHub/GitLab repository, and Google Colab notebook (if applicable) in the **Project Artifacts** section of the final report.
- You are advised to make your work clear and well presented; marks may be reduced for poor presentation. This includes filling in your information on the cover page.
- You MUST show all your work, and text **must not** be converted into an image unless specified otherwise by the question.

Task 4 – Reflection (2 Marks)

- **Late submission will result in ZERO marks being awarded**.
- The work should be your own. Copying **from students or other resources, including AI software, will result in ZERO marks**.

Learning Outcome(s):

CLO1, CLO2, CLO3, CLO4, CLO5

Project Description

Design and Implementation of a Smart Clinic Database System

A private clinic currently manages its patient information manually, making it difficult to organize appointments, treatments, and payments. The clinic has decided to develop a database system to improve data organization and retrieval.

Your team is required to design, implement, and test a complete relational database using the concepts learned throughout this course. The project should demonstrate your understanding of database design, ER/EER modeling, relational mapping, SQL implementation, and advanced SQL queries. The project must be completed using MySQL.

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Task 4 – Reflection (2 Marks)

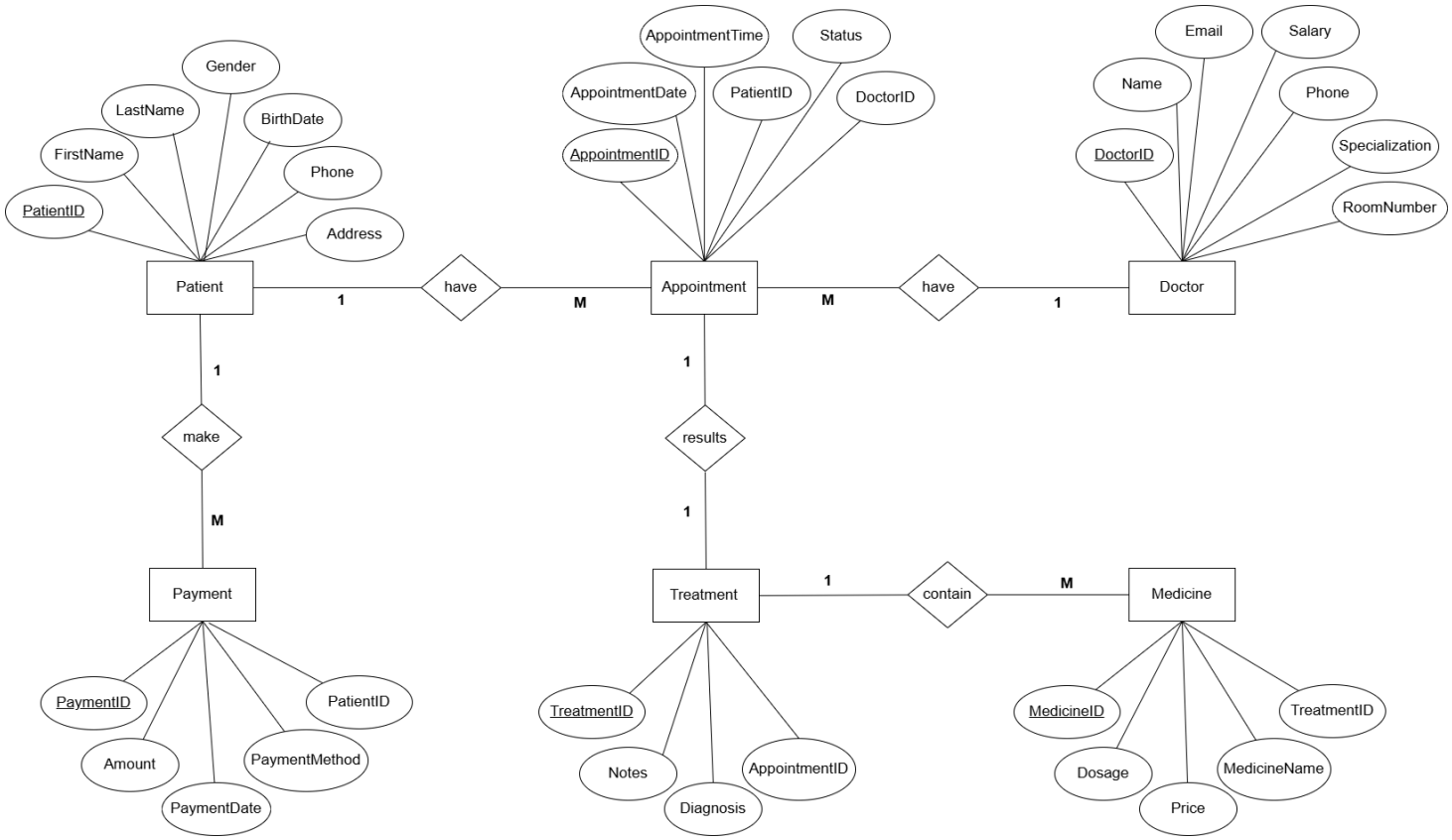
[redacted]

Task 1 – Database Design (6 Marks)

- Design a complete database for the clinic that includes at least six entities such as Patients, Doctors, Appointments, Treatments, Medicines, Payments, or any other appropriate entities.
- A complete ER/EER diagram.
- Primary and foreign keys.
- Appropriate relationships with cardinality.
- At least one specialization/generalization (EER feature).
- Any assumptions made during the design process.

Answer:

ER diagram:



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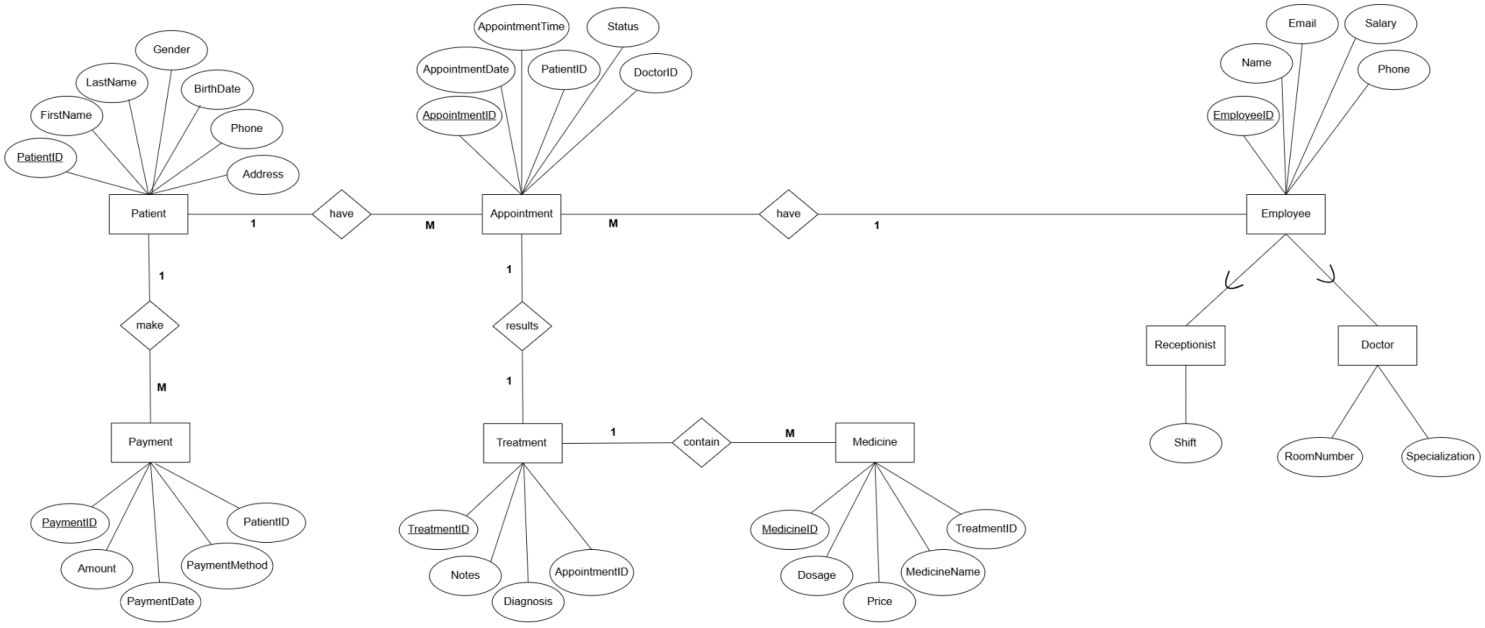
Task 4 – Reflection (2 Marks)

Table	Primary Key	Foreign Key
Patient	PatientID	-
Doctor	DoctorID	DoctorID
Appointment	AppointmentID	PatientID, DoctorID
Treatment	TreatmentID	AppointmentID
Medicine	MedicineID	TreatmentID
Payment	PaymentID	PatientID

EER diagram:



Assumptions:

- Each patient can have multiple appointments
- Each appointment belongs to one patient
- Each appointment is handled by one doctor
- Each treatment is linked to one appointment

Task 4 – Reflection (2 Marks)

- A treatment may include multiple medicines
- A patient may make multiple payments
- Every doctor and receptionist is an employee

Task 2 – Database Implementation (6 Marks)

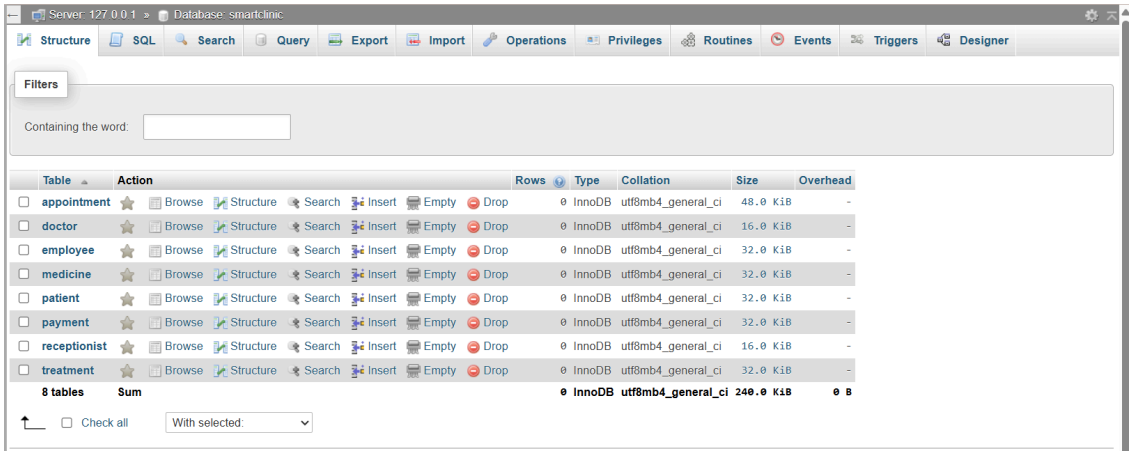
Complete the following tasks using MySQL. Include screenshots of both your SQL code and the corresponding output (results) for each task:

- Create the database and all required tables.
- Use appropriate data types and constraints (PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, etc.).
- Insert at least 5 records into each main table.

Answer:

CREATE DATABASE SmartClinic;

Screenshot:



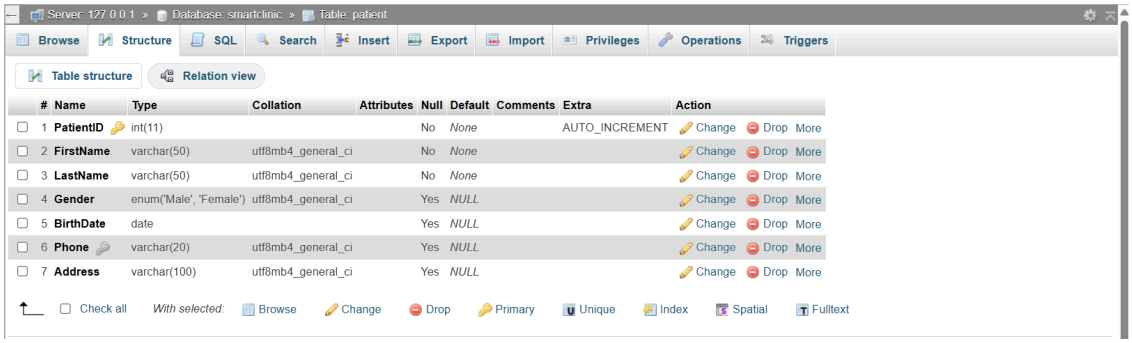

```
CREATE TABLE Patient (
    PatientID INT AUTO_INCREMENT PRIMARY KEY,
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50) NOT NULL,
    Gender ENUM('Male','Female'),
    BirthDate DATE,
    Phone VARCHAR(20) UNIQUE,
```

Task 4 – Reflection (2 Marks)

[redacted]

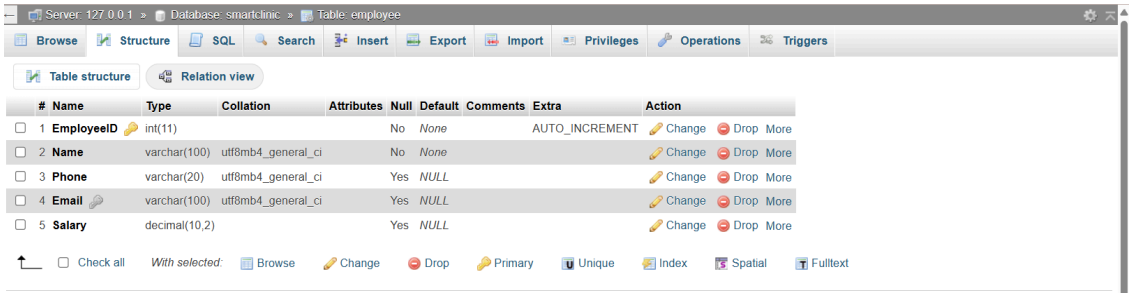
```
    Address VARCHAR(100)
);
```

Screenshot:




```
CREATE TABLE Employee (
    EmployeeID INT AUTO_INCREMENT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(20),
    Email VARCHAR(100) UNIQUE,
    Salary DECIMAL(10,2)
);
```

Screenshot:

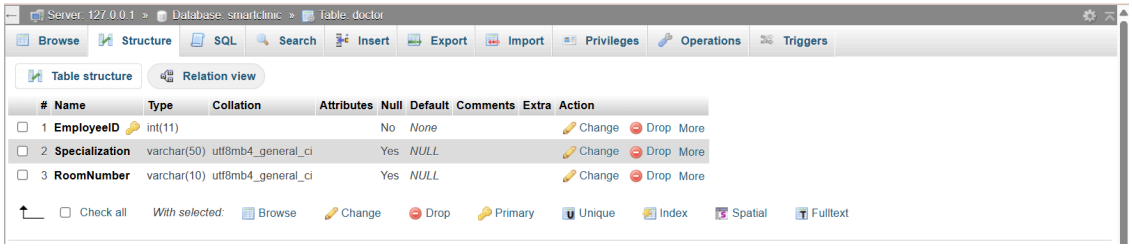



```
CREATE TABLE Doctor (
    EmployeeID INT PRIMARY KEY,
    Specialization VARCHAR(50),
```

Task 4 – Reflection (2 Marks)

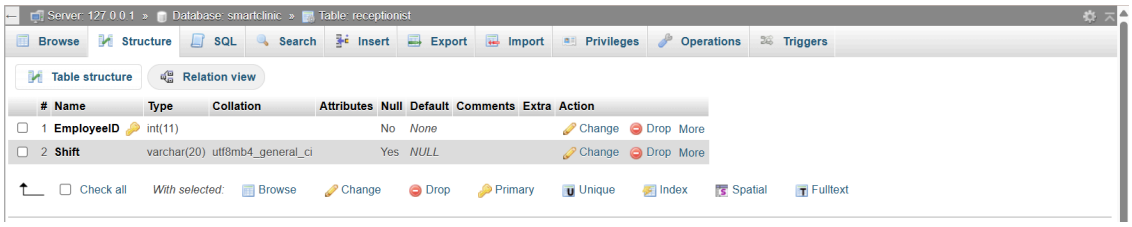
```
    RoomNumber VARCHAR(10),
    FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID)
);
```

Screenshot:




```
CREATE TABLE Receptionist (
    EmployeeID INT PRIMARY KEY,
    Shift VARCHAR(20),
    FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID)
);
```

Screenshot:



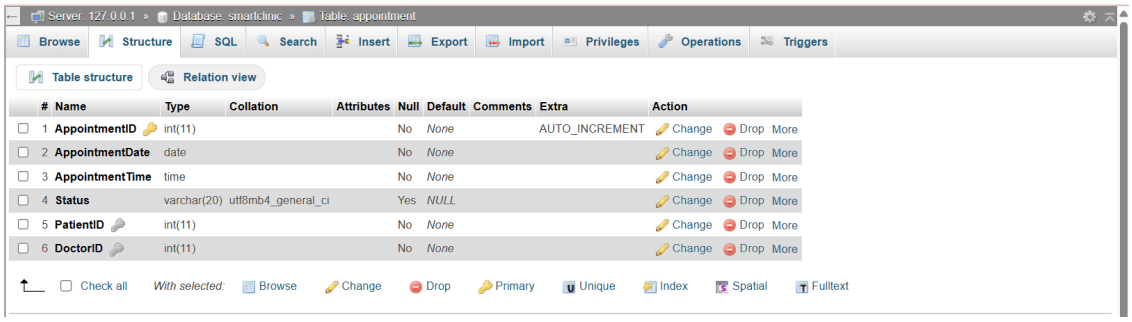

```
CREATE TABLE Appointment (
    AppointmentID INT AUTO_INCREMENT PRIMARY KEY,
    AppointmentDate DATE NOT NULL,
    AppointmentTime TIME NOT NULL,
    Status VARCHAR(20),
    PatientID INT NOT NULL,
    DoctorID INT NOT NULL,
    FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),
```

Task 4 – Reflection (2 Marks)

[redacted]

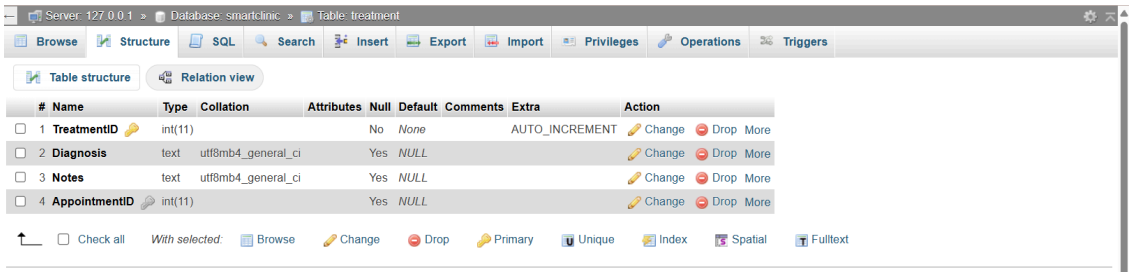
```
    FOREIGN KEY (DoctorID) REFERENCES Doctor(EmployeeID)
);
```

Screenshot:




```
CREATE TABLE Treatment (
    TreatmentID INT AUTO_INCREMENT PRIMARY KEY,
    Diagnosis TEXT,
    Notes TEXT,
    AppointmentID INT UNIQUE,
    FOREIGN KEY (AppointmentID) REFERENCES Appointment(AppointmentID)
);
```

Screenshot:



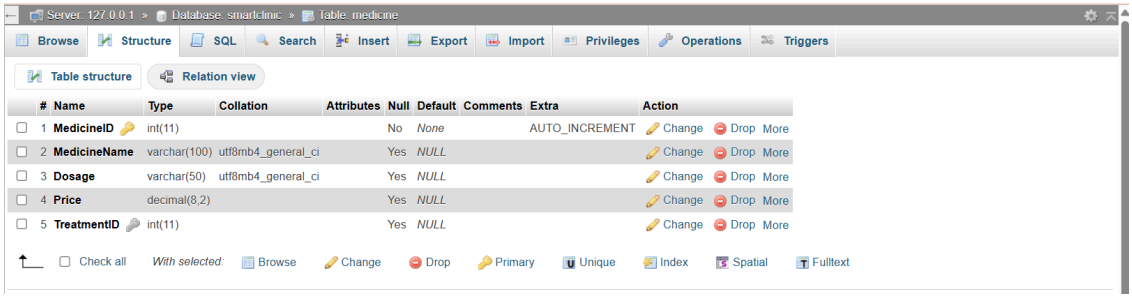

```
CREATE TABLE Medicine (
    MedicineID INT AUTO_INCREMENT PRIMARY KEY,
    MedicineName VARCHAR(100),
    Dosage VARCHAR(50),
    Price DECIMAL(8,2),
```

Task 4 – Reflection (2 Marks)

[redacted]

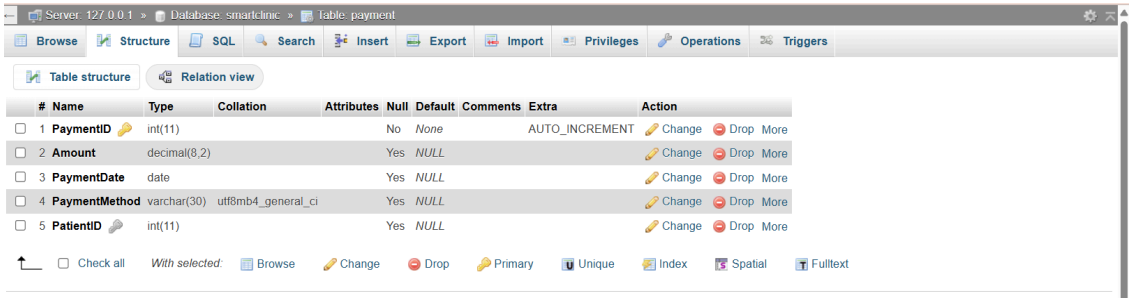
```
    TreatmentID INT,
    FOREIGN KEY (TreatmentID) REFERENCES Treatment(TreatmentID)
);
```

Screenshot:




```
CREATE TABLE Payment (
    PaymentID INT AUTO_INCREMENT PRIMARY KEY,
    Amount DECIMAL(8,2),
    PaymentDate DATE,
    PaymentMethod VARCHAR(30),
    PatientID INT,
    FOREIGN KEY (PatientID) REFERENCES Patient(PatientID)
);
```

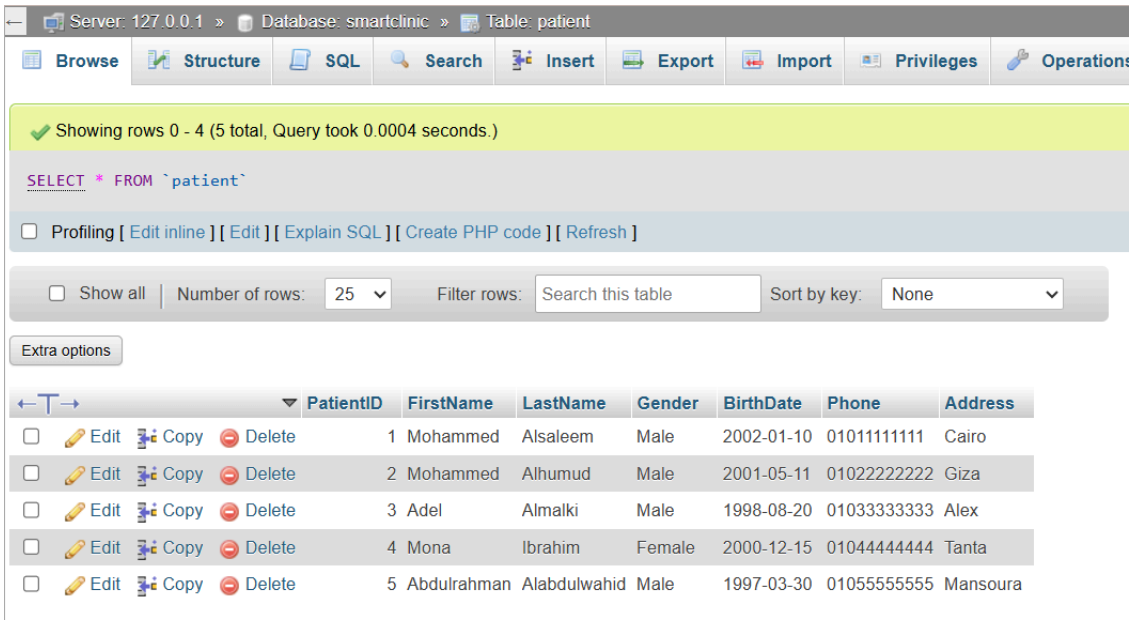
Screenshot:



Task 4 – Reflection (2 Marks)

INSERT INTO Patient(FirstName,LastName,Gender,BirthDate,Phone,Address) VALUES
('Mohammed','Alsaleem','Male','2002-01-10','01011111111','Cairo'),
('Mohammed','Alhumud','Male','2001-05-11','01022222222','Giza'),
('Adel','Almalki','Male','1998-08-20','01033333333','Alex'),
('Mona','Ibrahim','Female','2000-12-15','01044444444','Tanta'),
('Abdulrahman','Alabdulwahid','Male','1997-03-30','01055555555','Mansoura');

Screenshot:



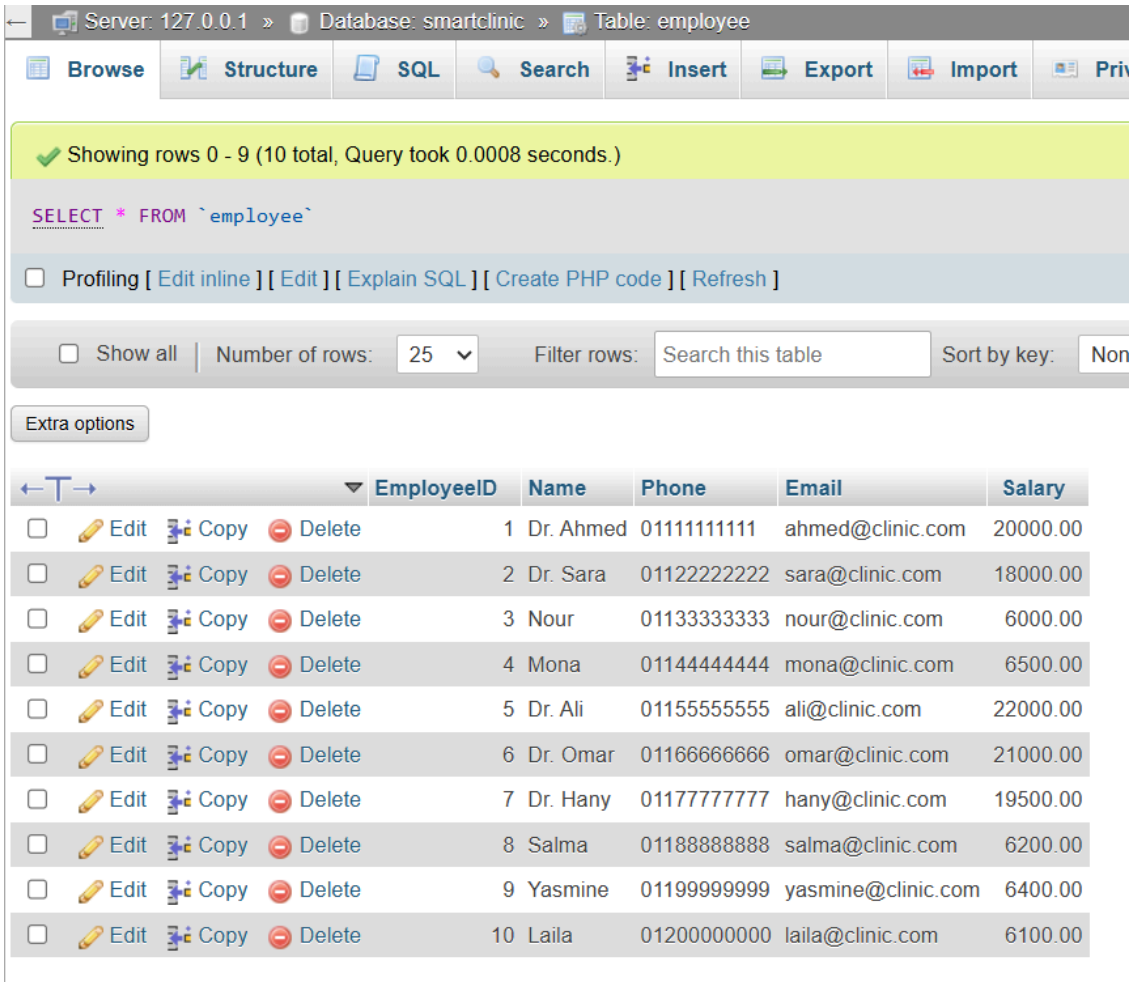
INSERT INTO Employee(Name,Phone,Email,Salary) VALUES
('Dr. Ahmed', '01111111111', 'ahmed@clinic.com', 20000),
('Dr. Sara', '01122222222', 'sara@clinic.com', 18000),
('Nour', '01133333333', 'nour@clinic.com', 6000),
('Mona', '01144444444', 'mona@clinic.com', 6500),
('Dr. Ali', '01155555555', 'ali@clinic.com', 22000),
('Dr. Omar', '01166666666', 'omar@clinic.com', 21000),
('Dr. Hany', '01177777777', 'hany@clinic.com', 19500),

Task 4 – Reflection (2 Marks)

[redacted]

('Salma', '01188888888', 'salma@clinic.com', 6200),
('Yasmine', '01199999999', 'yasmine@clinic.com', 6400),
('Laila', '01200000000', 'laila@clinic.com', 6100);

Screenshot:

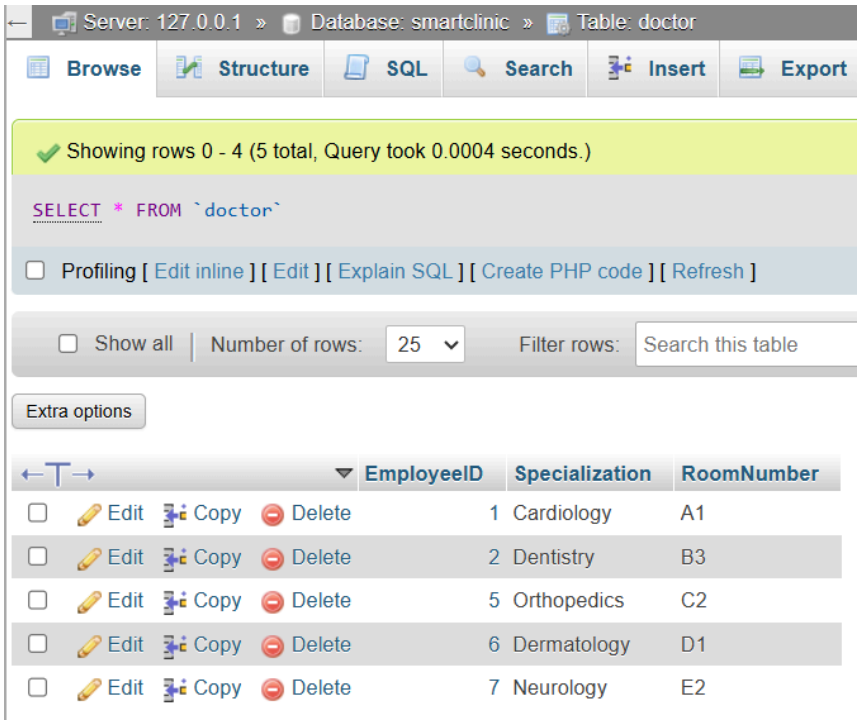


EmployeeID	Name	Phone	Email	Salary
1	Dr. Ahmed	01111111111	ahmed@clinic.com	20000.00
2	Dr. Sara	01122222222	sara@clinic.com	18000.00
3	Nour	01133333333	nour@clinic.com	6000.00
4	Mona	01144444444	mona@clinic.com	6500.00
5	Dr. Ali	01155555555	ali@clinic.com	22000.00
6	Dr. Omar	01166666666	omar@clinic.com	21000.00
7	Dr. Hany	01177777777	hany@clinic.com	19500.00
8	Salma	01188888888	salma@clinic.com	6200.00
9	Yasmine	01199999999	yasmine@clinic.com	6400.00
10	Laila	01200000000	laila@clinic.com	6100.00

INSERT INTO Doctor VALUES
(1, 'Cardiology', 'A1'),
(2, 'Dentistry', 'B3'),
(5, 'Orthopedics', 'C2'),
(6, 'Dermatology', 'D1'),

(7, 'Neurology', 'E2');

Screenshot:



INSERT INTO Receptionist VALUES

(3, 'Morning'),

(4, 'Evening'),

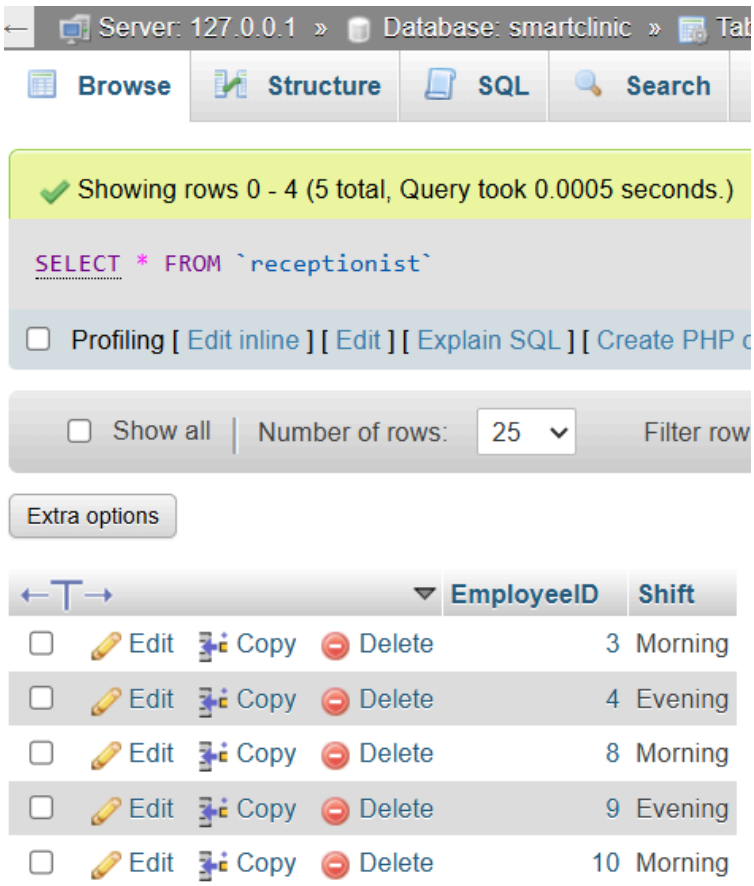
(8, 'Morning'),

(9, 'Evening'),

(10, 'Morning');

Screenshot:

Task 4 – Reflection (2 Marks)



INSERT INTO Appointment
(AppointmentDate,AppointmentTime,Status,PatientID,DoctorID) VALUES
('2026-08-01', '10:00:00', 'Completed', 1, 1),
('2026-08-01', '11:00:00', 'Completed', 2, 2),
('2026-08-02', '09:30:00', 'Pending', 3, 5),
('2026-08-03', '12:00:00', 'Completed', 4, 6),
('2026-08-04', '01:00:00', 'Pending', 5, 7);

Screenshot:

Task 4 – Reflection (2 Marks)

Server: 127.0.0.1 » Database: smartclinic » Table: appointment

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operat

Showing rows 0 - 4 (5 total, Query took 0.0004 seconds.)

```
SELECT * FROM `appointment`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

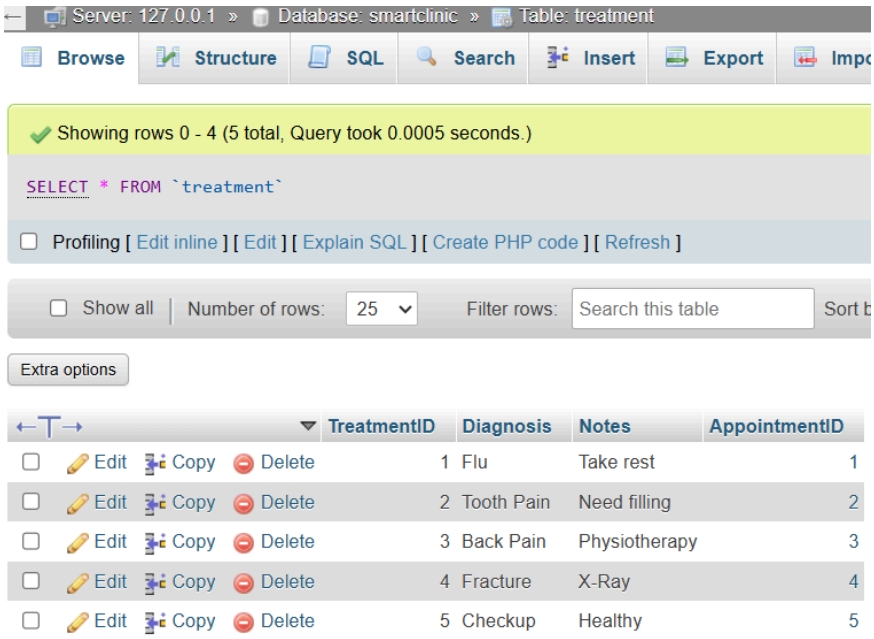
Extra options

	AppointmentID	AppointmentDate	AppointmentTime	Status	PatientID	DoctorID
Edit Copy Delete	1	2026-08-01	10:00:00	Completed	1	1
Edit Copy Delete	2	2026-08-01	11:00:00	Completed	2	2
Edit Copy Delete	3	2026-08-02	09:30:00	Pending	3	5
Edit Copy Delete	4	2026-08-03	12:00:00	Completed	4	6
Edit Copy Delete	5	2026-08-04	01:00:00	Pending	5	7

INSERT INTO Treatment(Diagnosis,Notes,AppointmentID) VALUES
('Flu','Take rest',1),
('Tooth Pain','Need filling',2),
('Back Pain','Physiotherapy',3),
('Fracture','X-Ray',4),
('Checkup','Healthy',5);

Screenshot:

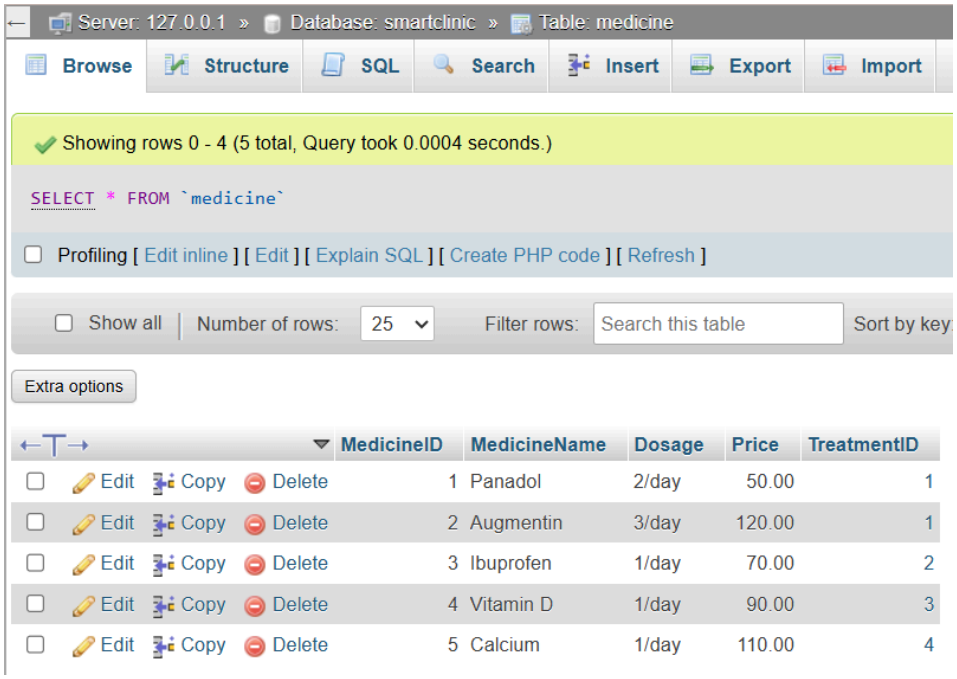
Task 4 – Reflection (2 Marks)



INSERT INTO Medicine(MedicineName,Dosage,Price,TreatmentID) VALUES
('Panadol','2/day',50,1),
('Augmentin','3/day',120,1),
('Ibuprofen','1/day',70,2),
('Vitamin D','1/day',90,3),
('Calcium','1/day',110,4);

Screenshot:

Task 4 – Reflection (2 Marks)



INSERT INTO Payment(Amount,PaymentDate,PaymentMethod,PatientID) VALUES
(300,'2026-08-01','Cash',1),
(450,'2026-08-01','Visa',2),
(350,'2026-08-02','Cash',3),
(700,'2026-08-03','Visa',4),
(250,'2026-08-04','Cash',5);

Screenshot:

Task 4 – Reflection (2 Marks)

[redacted]

Server: 127.0.0.1 » Database: smartclinic » Table: payment

Browse | Structure | SQL | Search | Insert | Export | Import | Pr

Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)

```
SELECT * FROM `payment`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: No

Extra options

←T→	PaymentID	Amount	PaymentDate	PaymentMethod	PatientID
Edit Copy Delete	1	300.00	2026-08-01	Cash	1
Edit Copy Delete	2	450.00	2026-08-01	Visa	2
Edit Copy Delete	3	350.00	2026-08-02	Cash	3
Edit Copy Delete	4	700.00	2026-08-03	Visa	4
Edit Copy Delete	5	250.00	2026-08-04	Cash	5

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Task 3 – SQL Operations (6 Marks)

Complete the following tasks using MySQL. For each task, include screenshots of both your SQL code and the corresponding output (results), along with a brief description (one to two sentences) explaining the purpose of the query and the information it is intended to retrieve.

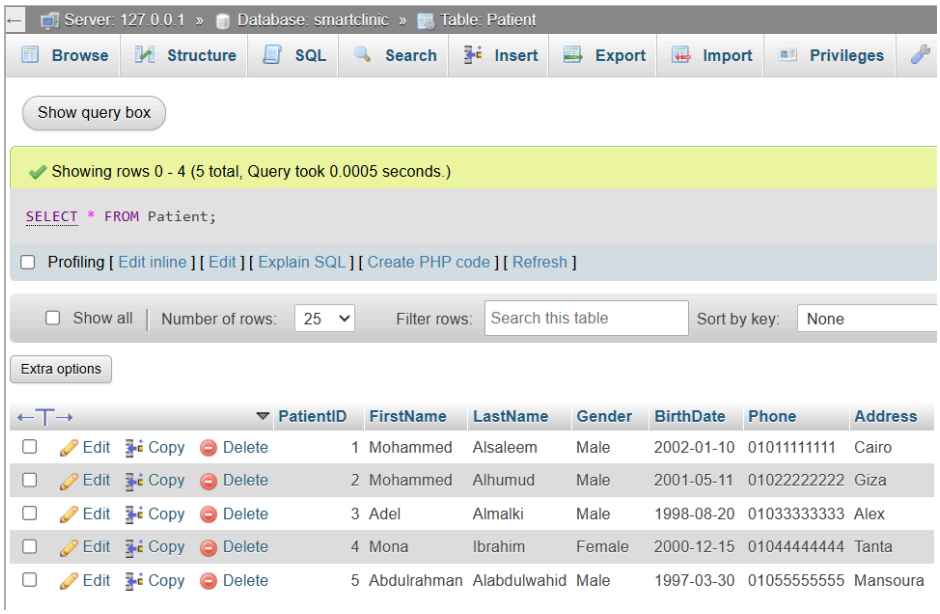
- Write SELECT statements.
- Write JOIN queries.
- Write nested queries.
- Use aggregate functions with GROUP BY.
- Write UPDATE and DELETE statements.
- Create one VIEW.
- Create one TRIGGER.

Answer:

-- SELECT statement

SELECT * FROM Patient;

Screenshot:



	PatientID	FirstName	LastName	Gender	BirthDate	Phone	Address
Edit Copy Delete	1	Mohammed	Alsaleem	Male	2002-01-10	01011111111	Cairo
Edit Copy Delete	2	Mohammed	Alhumud	Male	2001-05-11	01022222222	Giza
Edit Copy Delete	3	Adel	Almalki	Male	1998-08-20	01033333333	Alex
Edit Copy Delete	4	Mona	Ibrahim	Female	2000-12-15	01044444444	Tanta
Edit Copy Delete	5	Abdulrahman	Alabdulwahid	Male	1997-03-30	01055555555	Mansoura

Task 4 – Reflection (2 Marks)

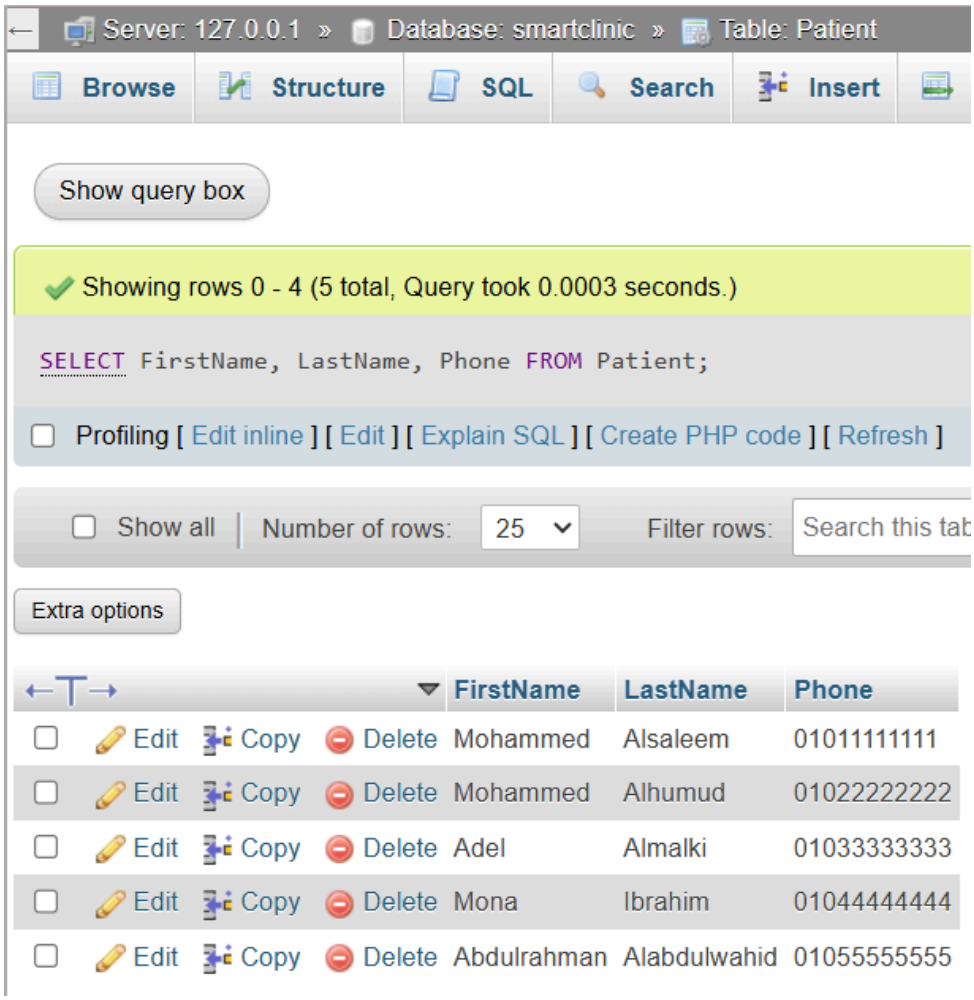
Description:

This query retrieves all records from the Patient table to display patient information

-- Another SELECT statement

SELECT FirstName, LastName, Phone FROM Patient;

Screenshot:



Description:

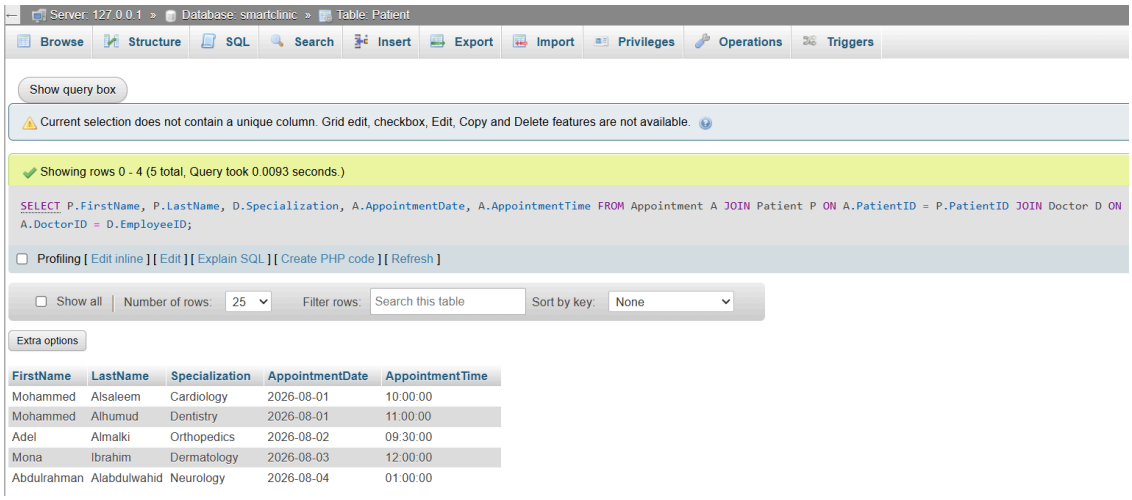
This query displays the patient's first name and last name and phone number only

Task 4 – Reflection (2 Marks)

-- JOIN query

```
SELECT
P.FirstName,
P.LastName,
D.Specialization,
A.AppointmentDate,
A.AppointmentTime
FROM Appointment A
JOIN Patient P ON A.PatientID = P.PatientID
JOIN Doctor D ON A.DoctorID = D.EmployeeID;
```

Screenshot:



FirstName	LastName	Specialization	AppointmentDate	AppointmentTime
Mohammed	Alsaleem	Cardiology	2026-08-01	10:00:00
Mohammed	Alhumud	Dentistry	2026-08-01	11:00:00
Adel	Almalki	Orthopedics	2026-08-02	09:30:00
Mona	Ibrahim	Dermatology	2026-08-03	12:00:00
Abdulrahman	Alabdulwahid	Neurology	2026-08-04	01:00:00

Description:

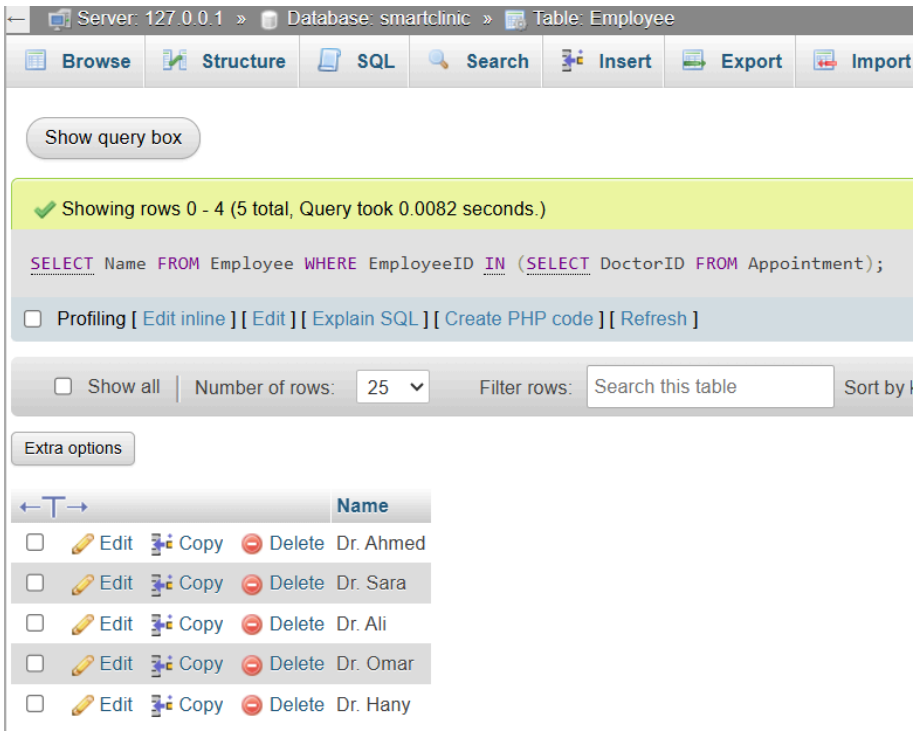
This query joins the Patient's first name and Patient's last name from Patient table and Appointment date and time from Appointment table and Doctor's specialization from Doctor table to display appointment details along with patient and doctor information

-- Nested query

```
SELECT Name FROM Employee WHERE EmployeeID
IN (SELECT DoctorID FROM Appointment);
```

Screenshot:

Task 4 – Reflection (2 Marks)



Description:

This query retrieves the names of doctors who have scheduled appointments

-- Aggregate function with GROUP BY

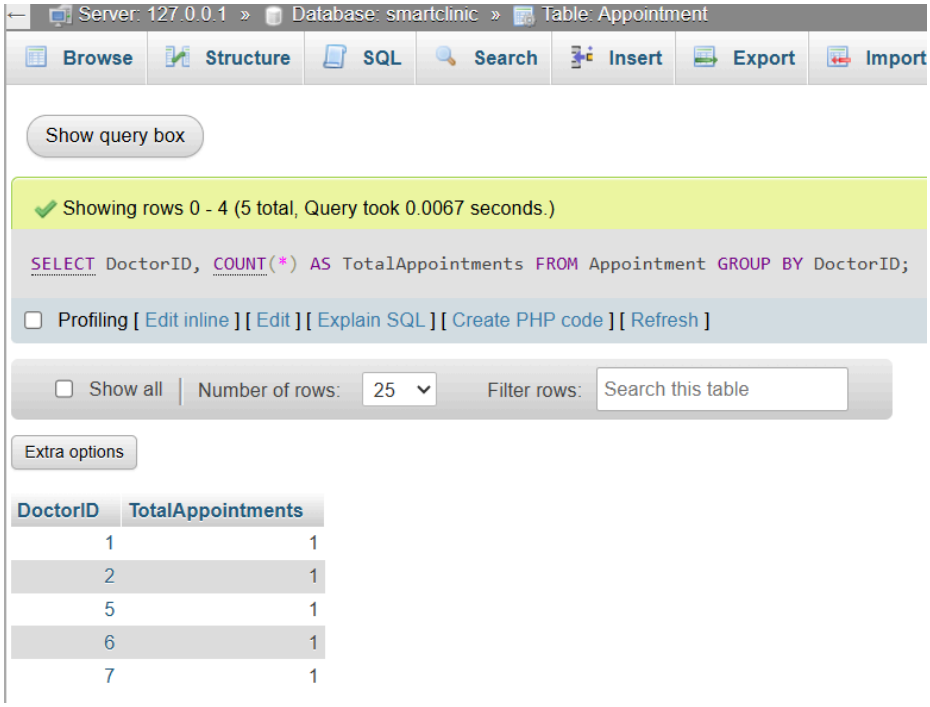
SELECT DoctorID, COUNT(*) AS TotalAppointments

FROM Appointment GROUP BY DoctorID;

Screenshot:

Task 4 – Reflection (2 Marks)

[redacted]



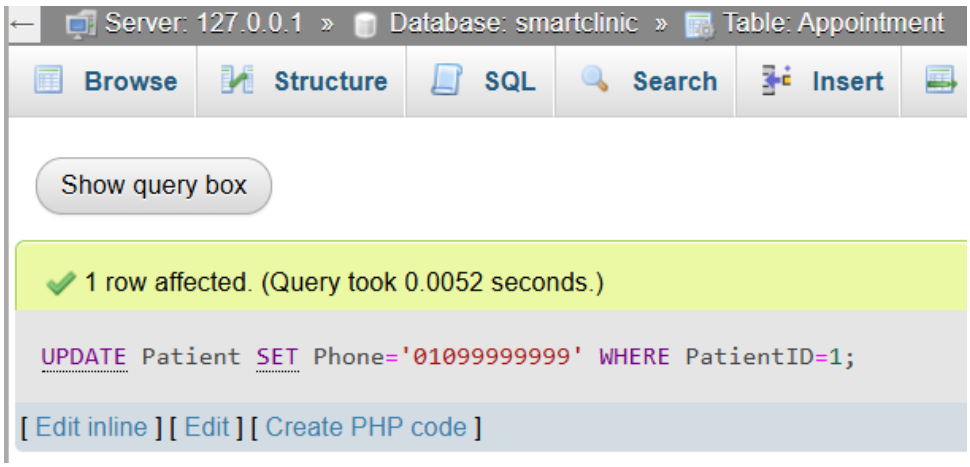
Description:

This query counts the total number of appointments assigned to each doctor

-- UPDATE statement

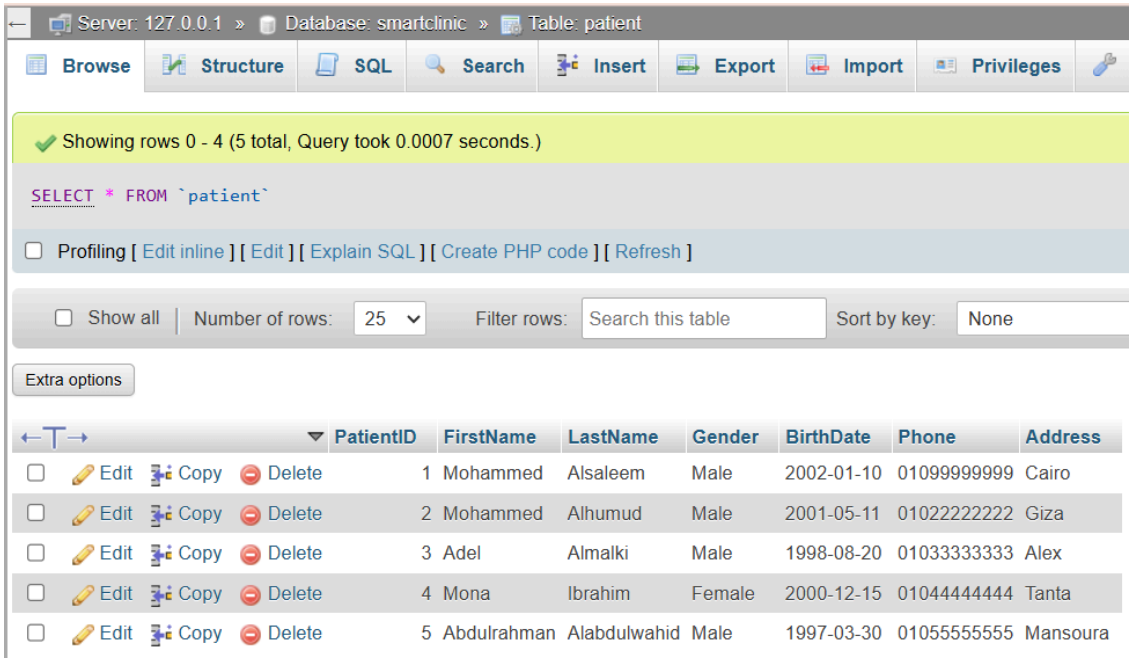
UPDATE Patient SET Phone='01099999999' WHERE PatientID=1;

Screenshot:



Task 4 – Reflection (2 Marks)

[redacted]



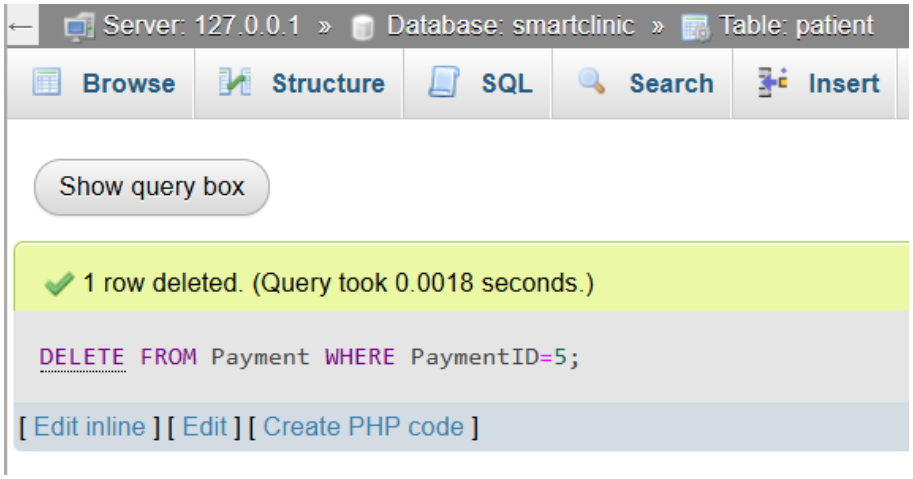
Description:

This query updates the phone number of a specific patient that his PatientID is 1

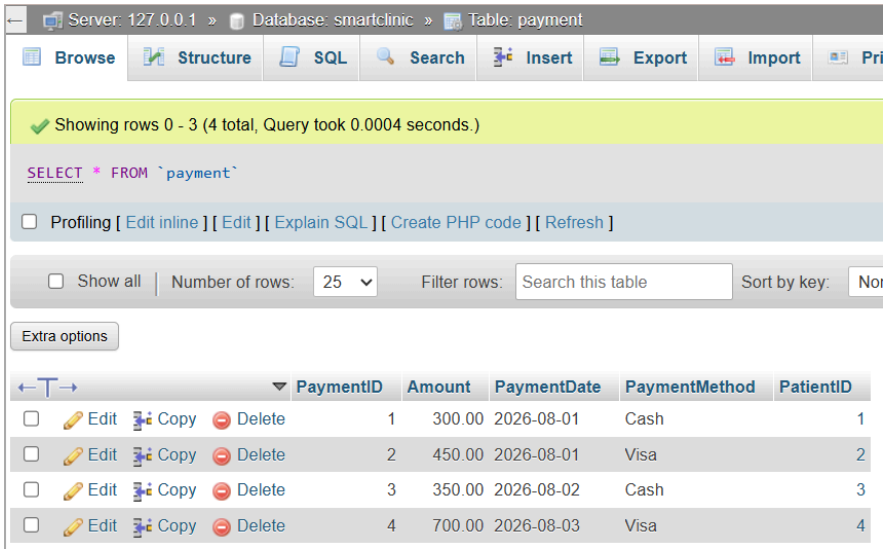
-- DELETE statement

DELETE FROM Payment WHERE PaymentID=5;

Screenshot:



Task 4 – Reflection (2 Marks)



Description:

This query removes a payment record from the Payment table that his PaymentID is 5

-- Create one VIEW

```
CREATE VIEW PatientAppointments AS
SELECT
P.FirstName,
P.LastName,
A.AppointmentDate,
A.AppointmentTime,
A.Status
FROM Patient P
JOIN Appointment A ON P.PatientID=A.PatientID;
```

Screenshot:

Task 4 – Reflection (2 Marks)

[redacted]

Server: 127.0.0.1 » Database: smartclinic » View: patientappointments

Browse | Structure | SQL | Search | Insert | Export | Privileges | Operatio

Current selection does not contain a unique column. Grid edit, Edit, Copy and Delete features may result in undesired beh

Showing rows 0 - 4 (5 total, Query took 0.0007 seconds.)

```
SELECT * FROM `patientappointments`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

				FirstName	LastName	AppointmentDate	AppointmentTime	Status
Edit	Copy	Delete		Mohammed	Alsaleem	2026-08-01	10:00:00	Completed
Edit	Copy	Delete		Mohammed	Alhumud	2026-08-01	11:00:00	Completed
Edit	Copy	Delete		Adel	Almalki	2026-08-02	09:30:00	Pending
Edit	Copy	Delete		Mona	Ibrahim	2026-08-03	12:00:00	Completed
Edit	Copy	Delete		Abdulrahman	Alabdulwahid	2026-08-04	01:00:00	Pending

Server: 127.0.0.1 » Database: smartclinic » Table: payment

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | Triggers

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0049 seconds.)

```
CREATE VIEW PatientAppointments AS SELECT P.FirstName, P.LastName, A.AppointmentDate, A.AppointmentTime, A.Status FROM Patient P JOIN Appointment A ON P.PatientID=A.PatientID;
```

[Edit inline] [Edit] [Create PHP code]

Description:

This view displays patient names together with their appointment details

-- Create one TRIGGER

DELIMITER $$

CREATE TRIGGER CheckPaymentAmount

BEFORE INSERT ON Payment

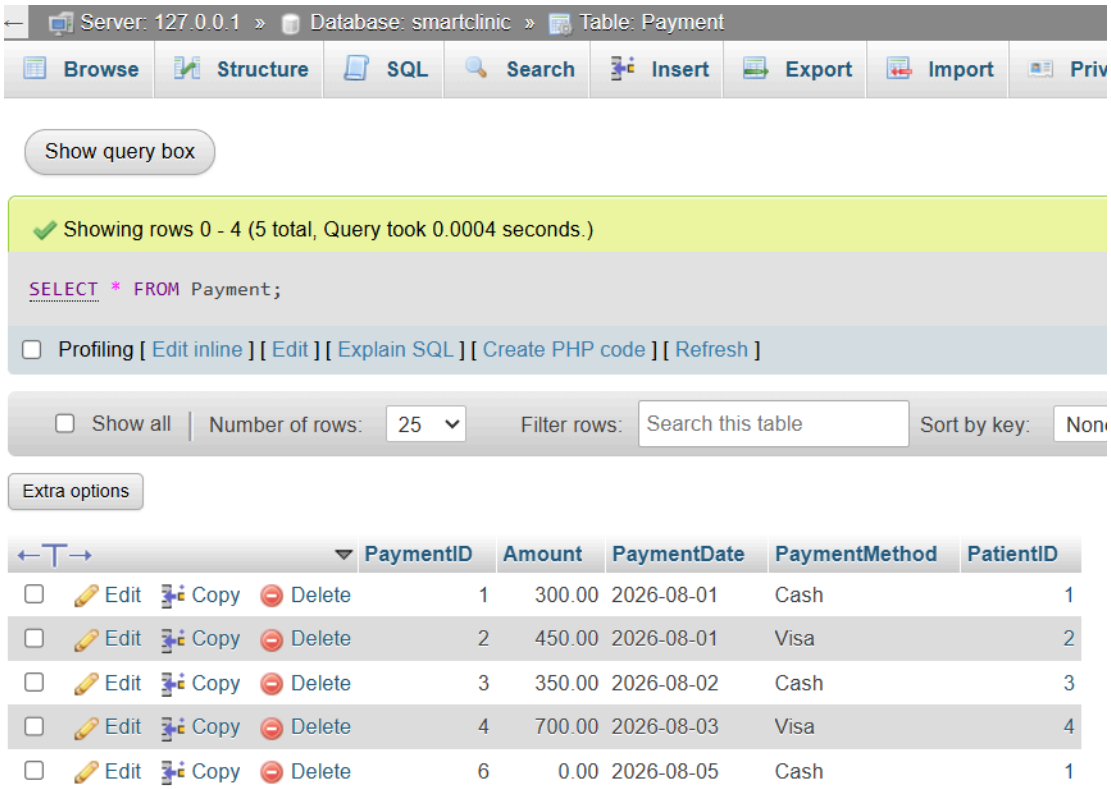
FOR EACH ROW

Task 4 – Reflection (2 Marks)

[redacted]

```
BEGIN
IF NEW.Amount < 0 THEN
SET NEW.Amount = 0;
END IF;
END $$
DELIMITER ;
```

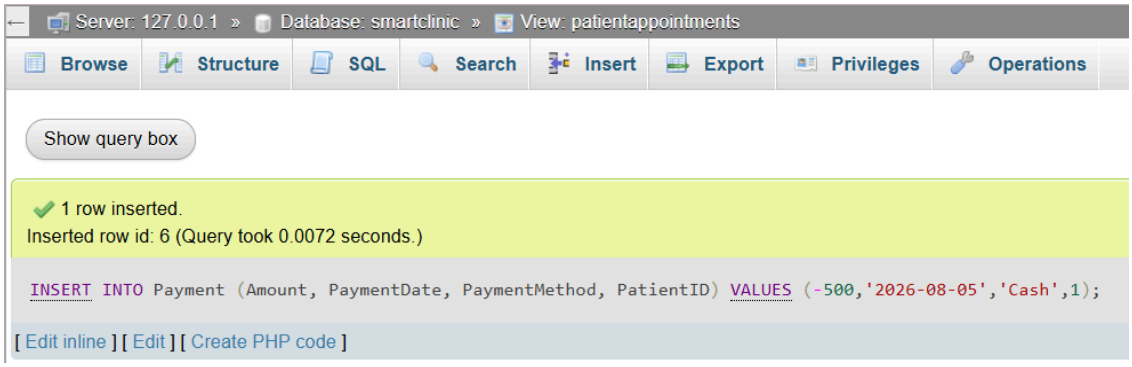
Screenshot:

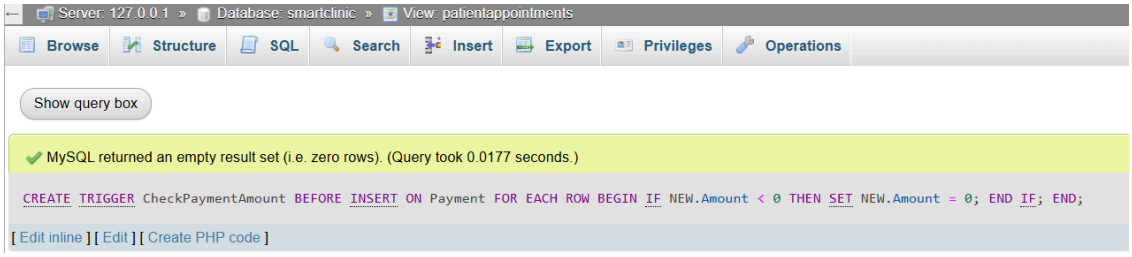


	PaymentID	Amount	PaymentDate	PaymentMethod	PatientID
Edit Copy Delete	1	300.00	2026-08-01	Cash	1
Edit Copy Delete	2	450.00	2026-08-01	Visa	2
Edit Copy Delete	3	350.00	2026-08-02	Cash	3
Edit Copy Delete	4	700.00	2026-08-03	Visa	4
Edit Copy Delete	6	0.00	2026-08-05	Cash	1

Task 4 – Reflection (2 Marks)

[redacted]





Description:

This trigger validates the payment amount before inserting a new record if the amount is negative it automatically replaces it with 0 to maintain data integrity and make example to insert negative amount

Restricted - مقيد

Task 4 – Reflection (2 Marks)

Write a **300–500-word reflection** on your experience while completing this project by answering the following questions. Your responses should be clear, concise, and supported with specific examples from your own work.

1. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
2. Explain why you chose your specific approach/method/model over at least one alternative you considered, and what tradeoffs informed that decision.
3. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
4. What would you do differently if you were to redo this project with more time or resources?

Answer:

Completing the Smart Clinic Database project was a valuable learning experience that improved my understanding of database design and SQL implementation on of the biggest challenges i faced was designing the relationships between the tables while maintaining proper normalization at first i was unsure how to connect the Employee and Doctor and Receptionist entities without creating redundant data after reviewing the project requirements and analyzing the relationships, we decided to use specialization in the EER model where Employee became the superclass and Doctor and Receptionist became subclasses this approach reduced data duplication and made the database structure more organized.

When designing the database i considered two different approaches the first approach was to create completely separate tables for doctors and receptionists each containing all employee information the second approach was to use a superclass Employee with specialized subclasses i chose the second approach because it follows good database design principles and avoids repeating common attributes such as name and phone and number and email and salary Although this method required additional foreign key

Restricted - مقيد

Task 4 – Reflection (2 Marks)

relationships and slightly increased the complexity of the design it provided better scalability and easier maintenance if new employee types were added in the future.

We didn't use any AI tools while completing this project. All aspects of the database design and SQL and implementation were completed independently.

If i had more time and resources we would expand the project by adding more advanced features for example we would create additional tables for departments and prescriptions and medical records we would also implement stored procedures more complex triggers and additional SQL views to improve the functionality of the system we would spend more time testing different scenarios and improving the user documentation overall this project strengthened my practical knowledge of database design and SQL programming and the importance of careful planning before implementation

Task 4 – Reflection (2 Marks)

Deliverables

1. **Project Report (PDF)** – System description, ER/EER diagram, relational schema, SQL code, screenshots, and reflection.
2. **SQL Script (.sql)** – Complete SQL implementation of the project.
3. **GitHub Repository** – Repository link with the complete project and commit history.
4. **Mid-Project Progress Report** – Summary of progress, challenges, decisions, and team contributions.
5. **Project Artifacts**
    - Live report document (with edit history): [link]
    - GitHub/GitLab repository (with commit history): [link]
    - Google Colab notebook (if applicable): [link]

Live report document:
https://docs.google.com/document/d/1t8yur5ryPM2rHABlqd1xwSC6OzaoQex7-j6-s_HIZio/edit?usp=sharing

GitHub/GitLab repository:

https://github.com/alsaleemmohammed076-crypto/Smart-Clinic-Database.git